

Ubuntu va OS asoslari: to'liq mashqlar

P0–P6: topshiriqlar va qadamma-qadam yechim

Ubuntu Server 24.04 LTS

JDU Cyber Security / 2026 / ChatGPT tarjima qoralamasi

Ubuntu and OS Foundations — to'liq ko'rsatmalar bilan mashqlar

ChatGPT tarjimasi joriy yaponcha manbaga moslashtirilgan. Ona tilida gapiruvchi mutaxassis hali tekshirmagan; baholash shartlari farq qilsa, yaponcha asl matnga tayaning.

Qamrov: Guided Practice P0–P6. Muhit: Ubuntu Server 24.04 LTS.

Tartib: avval har bir P topshirig'ini o'qing. O'zingiz yechib ko'ring, so'ng qadamma-qadam yechim bilan amallar va sabablarini solishtiring. Shu ko'nikmaga tegishli Mission Mni mustaqil bajaring.

Avval Ubuntuga ulaning

AWS Academy Learner Lab dastlabki sozlamalari tugagach, AWS konsolida CloudShellni oching. CloudShellda o'z Ubuntu mashq serveringizga ulaning:

```
ssh jdu-ubuntu
```

P0–P5ni ulangan Ubuntuda bajaring. P6da CloudShell va Ubuntu amallari uchun joy har bir bosqichda ko'rsatiladi.

0. Umumiy qoidalar

P0–P6 mashq uchun. P0 — OSni kuzatishga bag'ishlangan boshlang'ich qadamma-qadam mashq. P1–P6 M1–M6dan boshqa `directory`, `user`, `group`, `service` va `port`lardan foydalanadi. M7 umumlashtiruvchi topshiriq; unga alohida mashq yo'q.

Har bir Pda avval topshiriq matni, keyin yechim namunasi beriladi. Topshiriqni avval mustaqil bajarish yoki yo'riqnoma bilan birga ishlash mumkin. P va M alohida baholanadi: Pdagi `PASS` Mni avtomatik `PASS` qilmaydi.

Ubuntuda quyidagini bajaring. Natija `ssm-user` bo'lsin:

```
id -un
```

Boshqa `user` ko'rinsa, bir marta `exit` qiling va qayta tekshiring:

```
exit  
id -un
```

Har bir bosqich boshida ish `directory`siga o'ting. O'tgach `pwd` bilan joyni tekshiring.

Ubuntu tomonidagi baholashni `ssm-user` sifatida bajaring. P6ning CloudShell tomonini CloudShellda tekshiring:

```
jdu-check P1
```

Baholash yakuniy holatni tekshiradi. Faqat kuzatish yoki oraliq amal har doim baholanmaydi. Natija o'qituvchi progress sahifasiga avtomatik yuboriladi. Yuborish ishlamasa ham Ubuntu yoki CloudShell'dagi

PASS/FAIL ko'rinadi.

Faqat boshidan boshlash zarur bo'lsa `reset` qiling:

```
jdu-reset P1
```

P0 Environment and OS

O'rganiladigan ko'nikmalar

- Ubuntu OS haqidagi ma'lumot bilan Linux `kernel` haqidagi ma'lumotni farqlash.
- Hozirgi `user`, `hostname` va `PID 1` `processini` haqiqiy serverda aniqlash.
- `Command` natijalarini `filega` yozish, saqlash, tekshirish va topshirish.

Topshiriq

O'z Ubuntu mashq serveringizdan oltita qiymatni aniqlang va `~/jdu-lab/p0/observation.env`ga yozing. Bu eski `M0` o'rniga qadamma-qadam `P0` mashqidir. Qiymatlarni misoldan yoki boshqa talabadan ko'chirmang; faqat o'z serveringizdagi natijalarni oling. Oltita bandni `jdu-check P0` bilan tekshiring.

Yechim namunasi (qadamma-qadam)

1-bosqich: Server va userni tekshirish

Hujjat boshida aytilganidek, CloudShellda `ssh jdu-ubuntu`ni bajaring. Quyidagi barcha amallar ulangan Ubuntu'da bajariladi.

```
# Home directoryga o'ting.  
cd ~  
# Joriy directoryni ko'rsating.  
pwd  
# Joriy user nomini ko'rsating.  
id -un
```

`id -un` natijasi `ssm-user` bo'lishi kerak. Bu keyin yoziladigan `USER_NAME` qiymati. Keyingi natijalarni o'zingiz uchun qayd eting.

2-bosqich: OS turi va versiyasini aniqlash

```
# OS turi va versiyasini ko'rsating.  
cat /etc/os-release
```

`ID=` qiymatini `OS_ID`, `VERSION_ID=` qiymatini `OS_VERSION_ID` sifatida yozing. Bular `NAME` va `PRETTY_NAME`dan boshqa maydonlar. Agar natijada qo'shtirnoq bo'lsa, faqat qiymatni yozing.

3-bosqich: Linux kernel versiyasini aniqlash

```
# Linux kernel release raqamini ko'rsating.  
uname -r
```

`-r` kernel releasesini ko'rsatadi. Chiqqan butun satrni `KERNEL_RELEASE` sifatida qayd eting. OS versiyasi va kernel versiyasi bir xil ma'lumot emas.

4-bosqich: PID 1 process nomini aniqlash

```
# PID 1 process nomini ko'rsating.
cat /proc/1/comm
```

/procda ishlayotgan tizim haqidagi ma'lumotlar mavjud. 1 — process ID (PID), comm esa shu process nomini ko'rsatadigan file. Natijani PID1_COMM deb yozing.

5-bosqich: User va hostname qiymatlarini tekshirish

```
# Joriy user nomini ko'rsating.
id -un
# Server hostname qiymatini ko'rsating.
hostname
```

Birinchi natija USER_NAME, ikkinchisi HOST_NAME. Hostname serverni bildiradi; u user nomi yoki natija yuborishga ishlatiladigan Server ID emas.

6-bosqich: Kuzatuv fileini yaratish

```
# Home directoryga o'ting.
cd ~
# P0 ish directoriesini yarating.
mkdir -p jdu-lab/p0
# P0 ish directoriesiga o'ting.
cd jdu-lab/p0
# Joriy directoryni ko'rsating.
pwd
# Kuzatuv fileini tahrirlang.
nano observation.env
```

nano ochilgach, quyidagi olti satrni kiriting. Har bir = belgisidan o'ng tomonga o'z serveringizda aniqlagan qiymatni yozing. Bu file mazmuni; uni command sifatida bajarmang:

```
OS_ID=
OS_VERSION_ID=
KERNEL_RELEASE=
PID1_COMM=
USER_NAME=
HOST_NAME=
```

Har bir nom bitta satrda bo'lsin. = atrofida bo'sh joy va qiymat atrofida qo'shtirnoq bo'lmasin. Ctrl+O bilan saqlang, observation.env nomini Enter bilan tasdiqlang, so'ng Ctrl+X bilan chiqing.

7-bosqich: Saqlangan fileni tekshirish

```
# Joriy directoryni ko'rsating.
pwd
# Kuzatuv filei mavjudligi va permissionini tekshiring.
ls -l observation.env
# Yozib olingan oltita maydonni ko'rsating.
cat observation.env
```

Olti satrning hammasida 1–5-bosqichlarda olingan qiymatlar bo'lsin. Bo'sh yoki noto'g'ri qiymatni nano observation.env bilan tuzatib, qayta saqlang.

8-bosqich: Baholash va topshirish

```
# Baholash qaysi user bilan bajarilishini tekshiring.  
id -un  
# P0 bajarilganini baholang.  
jdu-check P0
```

Oltita `PASS` va `6 / 6 checks cleared` `P0` tugaganini bildiradi. `FAIL` bo'lsa, ko'rsatilgan bandni o'z serveringizda qayta aniqlang, `fileni` tuzating va baholashni takrorlang.

`RESULT` — topshiriq bahosi, `REPORT` — o'qituvchiga yuborish natijasi. Yuborish ishlamasa, `fileni` o'chirmang va mashqni `reset` qilmang. Ulanish hamda ro'yxatdan o'tishni tekshirib, yana urinib ko'ring.

9-bosqich: Keyingi mashqqa o'tish

```
# Keyingi mashqdan oldin home directoryga qayting.  
cd ~
```

`P1`ga o'ting. CloudShellga qaytmoqchi bo'lsangizgina `exit`ni bajaring. `P0`ni birinchi marta bajarishda `reset` kerak emas. Yozuvni o'chirib qayta boshlash kerak bo'lsagina Ubuntuda `jdu-reset` `P0`ni bajaring.

P1 Shell, path, file, text

O'rganiladigan ko'nikmalar

- `home directory` va ish `directorysini` ajratish.
- `directory` yaratish.
- `file`dan nusxa olish.
- `log`dan kerakli satrlarni saqlash.

Topshiriq

`~/jdu-lab/p1/inbox`da sozlama `filei` va `log` bor. `~/jdu-lab/p1/practice01` hali tayyor emas, unda keraksiz `.tmp` `filelar` qolgan. Quyidagi holatga keltiring:

1. `practice01` ichida `config`, `logs`, `notes` `directorylarini` yarating.
2. `inbox/config/training.conf`ni `practice01/config/training.conf`ga, `inbox/logs/practice.log`ni `practice01/logs/practice.log`ga ko'chiring. Manba va nusxa mazmunini o'zgartirmang.
3. `practice01` ostidagi barcha `.tmp` `filelarni` o'chiring. Boshqa `filelarni` o'chirmang.
4. Tayyor `practice01` ichidagi `directory` va `filelarning` egasi Ubuntu boshqaruv `useri` `ssm-user` bo'lsin.
5. Nusxa olingan `log`dagi `WARN` bor satrlarnigina satr raqamisiz `practice01/notes/warnings.txt`ga yozing.
6. Shu `log`ning oxirgi to'rt satrini asl tartibda `practice01/notes/recent.txt`ga yozing.

Tayyor holatni `jdu-check` `P1` bilan tekshiring. Olti bandning hammasi `PASS` bo'lsa, mashq tugaydi.

Yechim namunasi (qadamma-qadam)

1-bosqich: Home directoryga o'tish

```
# Home directoryga o'ting.  
cd ~  
# Joriy directoryni ko'rsating.  
pwd
```

Natija `/home/ssm-user` bo'lsin.

2-bosqich: P1 directorysiga o'tish

```
# P1 ish directorysiga o'ting.  
cd ~/jdu-lab/p1  
# Joriy directoryni ko'rsating.  
pwd  
# Asl filelar va tugallanmagan topshirish joyini daraxt shaklida ko'rsating.  
tree
```

`inbox` va hali tugallanmagan `practice01`ni toping.

3-bosqich: Kerakli directorylarni yaratish

```
# P1 ish directorysiga o'ting.  
cd ~/jdu-lab/p1  
# Config file uchun directory yarating.  
mkdir -p practice01/config  
# Log uchun directory yarating.  
mkdir -p practice01/logs  
# Natija filei uchun directory yarating.  
mkdir -p practice01/notes  
# Yaratilgan tuzilishni ko'rsating.  
tree practice01
```

4-bosqich: Config filedan nusxa olish

Nusxa joylashtiriladigan `directory`ga o'ting.

```
# Config nusxasi joylashgan directoryga o'ting.  
cd ~/jdu-lab/p1/practice01/config  
# Joriy directoryni tekshiring.  
pwd  
# Asl config fileni joriy directoryga nusxalang.  
cp ../../inbox/config/training.conf .  
# Nusxalangan fileni ko'rsating.  
ls -l
```

`.` hozirgi `directory`ni anglatadi.

5-bosqich: Log filedan nusxa olish

```
# Log nusxasi joylashgan directoryga o'ting.  
cd ~/jdu-lab/p1/practice01/logs  
# Joriy directoryni tekshiring.  
pwd  
# Asl logni joriy directoryga nusxalang.  
cp ../../inbox/logs/practice.log .  
# Nusxalangan fileni ko'rsating.  
ls -l
```

6-bosqich: Vaqtinchalik filelarni topib o'chirish

practice01ga o'ting va o'chirishdan oldin tuzilmani ko'ring:

```
# Yakunlash kerak bo'lgan directoryga o'ting.
cd ~/jdu-lab/p1/practice01
# Joriy directoryni tekshiring.
pwd
# O'chirishdan oldingi file tuzilishini ko'rsating.
tree
```

staging ichidan training.conf.tmp va practice.log.tmpni toping. So'ng shu ikki fileni shart bilan qidiring:

```
# Joriy directory va uning ichidagi faqat .tmp filelarni qidiring.
find . -type f -name '*.tmp' -print
```

. — qidirish boshlanadigan joy; -type f faqat oddiy filelarni, -name '*.tmp' nomi .tmp bilan tugaydiganlarni tanlaydi; -print topilgan pathlarni ko'rsatadi. Qo'shtirnoq *.tmpni shell oldindan kengaytirib yubormasligi uchun kerak. Natijani tree bilan solishtiring. Faqat ikki keraksiz file ekanini aniqlagach, aynan ularning pathini ko'rsatib o'chiring:

```
# Tekshirilgan ikkita keraksiz vaqtinchalik fileni o'chiring.
rm staging/training.conf.tmp staging/practice.log.tmp
# O'chirishdan keyingi tuzilishni tekshiring.
tree
```

Ikki .tmp ko'rinmaydi, lekin config/training.conf va logs/practice.log saqlanadi.

7-bosqich: WARN bor satrlarni saqlash

Natija yoziladigan notesga o'ting:

```
# Natija filei directorysiga o'ting.
cd ~/jdu-lab/p1/practice01/notes
# Joriy directoryni tekshiring.
pwd
# Logdagi WARN bor satrlarni ekranda ko'rsating.
grep 'WARN' ../logs/practice.log
```

Faqat WARN bor satrlar ko'ringanini tekshiring. Endi > qo'shib, shu natijani filega yozing. > natijani ekrandan filega yo'naltiradi; mavjud mazmunni qayta yozadi:

```
# WARN satrlarini warnings.txt fileiga saqlang.
grep 'WARN' ../logs/practice.log > warnings.txt
# Saqlangan natijani ko'rsating.
cat warnings.txt
```

Saqlangan satrlarni oldin ekranda ko'ringan satrlar bilan solishtiring.

8-bosqich: Oxirgi to'rt satrni saqlash

```
# Natija filei directorysiga o'ting.
cd ~/jdu-lab/p1/practice01/notes
# Logning oxirgi to'rt satrini ekranda ko'rsating.
tail -n 4 ../logs/practice.log
```

Oxirgi to'rt satr asl tartibda ko'ringanini tekshiring. So'ng aynan shu natijani saqlang:

```
# Oxirgi to'rt satrni recent.txt fileiga saqlang.
tail -n 4 ../logs/practice.log > recent.txt
# Saqlangan natijani ko'rsating.
cat recent.txt
```

Saqlangan to'rt satrni `cat` bilan tekshiring.

9-bosqich: Barcha natijani tekshirish

```
# P1 ish directoriesiga qayting.
cd ~/jdu-lab/p1
# Tayyor directory tuzilishini ko'rsating.
tree practice01
# P1 bajarilganini baholang.
jdu-check P1
```

Barcha 6 band `PASS` bo'lsa, M1ga o'ting.

P2 User, group, permission, setgid

O'rganiladigan ko'nikmalar

- `primary group` va `supplementary group`ni tekshirish.
- Umumiy directoryga `group permission` qo'yish.
- `setgid` yordamida yangi filening `group`ini meros qildirish.

Topshiriq

`/srv/jdu-practice-share` umumiy directorysida yozuvchi va o'quvchi huquqlarini ajrating. Dastlab `jdupracticewriter` noto'g'ri ravishda `practiceops` `supplementary group`iga kirgan, `jdupracticewriter` esa kirmagan. Directory va `GUIDE.txt`ning egasi, `group`i va `permission`i ham tayyor emas.

1. `jdupracticewriter`ni `practiceops` `supplementary group`iga qo'shing, `jdupracticewriter`ni undan chiqaring.
2. `/srv/jdu-practice-share` egasi va `group`ini `root:practiceops`, `mode`ini `2775` qiling. Yangi filelar directory `group`ini meros qilsin.
3. `GUIDE.txt` egasi va `group`ini `root:practiceops`, `mode`ini `664` qiling.
4. `ssm-user`dan `jdupracticewriter`ga o'tib, umumiy directoryda `writer-created.txt` yarating. Yangi file `group`i `practiceops` ekanini tekshiring.
5. `ssm-user`ga qaytib, `jdupracticewriter`ga o'ting. `GUIDE.txt`ni o'qish mumkin, lekin umumiy directoryda yangi file yaratish mumkin emasligini tekshiring.

Boshqa `user` bilan amallar tugagach, `ssm-user`ga qayting va `jdu-check P2`ni bajaring. Oltita baho `group` a'zoli, directory, file, `group` merosi va o'quvchi huquqiga tegishli.

Yechim namunasi (qadamma-qadam)

1-bosqich: Boshqaruv userini tekshirish

```
# Home directoryga o'ting.  
cd ~  
# Joriy user nomini ko'rsating.  
id -un  
# Joriy user ID va tegishli grouplarni ko'rsating.  
id
```

2-bosqich: Boshlang'ich groupni tekshirish

```
# Umumiy group yozuvini ko'rsating.  
getent group practiceops  
# Writer user tegishli grouplarni tekshiring.  
id jdupracticewriter  
# Viewer user tegishli grouplarni tekshiring.  
id jdupracticeviewer
```

Dastlab `jdupracticeviewer` `practiceops` a'zosi, `jdupracticewriter` esa a'zo emas.

3-bosqich: Group a'zolicini to'g'rilash

```
# Home directoryga o'ting.  
cd ~  
# Writer userni umumiy groupga qo'shing.  
sudo usermod -aG practiceops jdupracticewriter  
# Viewer userni umumiy groupdan chiqaring.  
sudo gpasswd -d jdupracticeviewer practiceops  
# O'zgarishdan keyingi umumiy groupni ko'rsating.  
getent group practiceops  
# Writer userning group a'zolicini tekshiring.  
id jdupracticewriter  
# Viewer userning group a'zolicini tekshiring.  
id jdupracticeviewer
```

4-bosqich: Umumiy directoryni sozlash

```
# /srv directoriesiga o'ting.  
cd /srv  
# Joriy directoryni ko'rsating.  
pwd  
# Umumiy directoryning boshlang'ich permissionini ko'rsating.  
ls -ld jdu-practice-share
```

O'zgartirishdan oldin `directoryning` `permissioni`, egasi va `groupini` ko'ring. `ls -ld`dagi `-d` `directoryning` ichidagilarni emas, aynan o'zini ko'rsatadi. Avval `permission`, havola sonidan keyin esa `ega` va `group` chiqadi.

```
# Umumiy directory owner va group qiymatini o'zgartiring.  
sudo chown root:practiceops jdu-practice-share  
# Owner va group o'zgarganini tekshiring.  
ls -ld jdu-practice-share  
# Group merosini o'z ichiga olgan 2775 permissionni o'rnatish.  
sudo chmod 2775 jdu-practice-share  
# setgid uchun s belgisi va permission qiymatini tekshiring.  
ls -ld jdu-practice-share
```

chowndan keyin ega va group root practiceops, chmoddan keyin permission drwxrwsr-x bo'lsin. Groupning bajarish joyidagi s — setgid; shu directoryda yaratilgan yangi file uning groupini oladi.

5-bosqich: GUIDE.txtni sozlash

Kerakli directoryga o'ting:

```
# Umumiy directoryga o'ting.
cd /srv/jdu-practice-share
# Joriy directoryni ko'rsating.
pwd
# GUIDE.txtning boshlang'ich permission qiymatini ko'rsating.
ls -l GUIDE.txt
```

Avval filening boshlang'ich permissioni, egasi va groupini ko'ring:

```
# GUIDE.txt owner va group qiymatlarini o'zgartiring.
sudo chown root:practiceops GUIDE.txt
# Owner va groupni tekshiring.
ls -l GUIDE.txt
# Owner va groupga o'qish hamda yozish huquqini bering.
sudo chmod 664 GUIDE.txt
# Filening yakuniy permissionini tekshiring.
ls -l GUIDE.txt
```

Yakuniy natija root practiceops va -rw-rw-r--. Ega va group o'qib-yozadi, boshqa userlar faqat o'qiydi.

6-bosqich: Writerga o'tib file yaratish

```
# Home directoryga qayting.
cd ~
# Writer user shelliga o'ting.
sudo su - jdupracticewriter
# Almashgandan keyingi user nomini tekshiring.
id -un
# Almashgandan keyingi grouplarni tekshiring.
id
# Umumiy directoryga o'ting.
cd /srv/jdu-practice-share
# Joriy directoryni tekshiring.
pwd
# Writer user sifatida file yarating.
printf '%s\n' 'guided writer file' > writer-created.txt
# Yaratilgan filening owner groupini tekshiring.
ls -l writer-created.txt
# Writer user shellidan chiqing.
exit
# Boshqaruvchi userga qaytganingizni tekshiring.
id -un
```

Oxirgi natija ssm-user bo'lsin.

7-bosqich: Viewerning o'qish va yozishini tekshirish

```
# Home directoryga qayting.
cd ~
# Viewer user shelliga o'ting.
sudo su - jdupracticviewer
# Almashgandan keyingi user nomini tekshiring.
id -un
# Viewer user tegishli grouplarni tekshiring.
id
# Umumiy directoryga o'ting.
cd /srv/jdu-practice-share
# Joriy directoryni tekshiring.
pwd
# GUIDE.txtni viewer user sifatida o'qing.
cat GUIDE.txt
# Yozish rad etilishini tekshiring.
touch viewer-created.txt
# Viewer user shellidan chiqing.
exit
# Boshqaruvchi userga qaytganingizni tekshiring.
id -un
```

cat muvaffaqiyatli ishlaydi. touch Permission denied beradi. Bu kutilgan rad etish.

jdupracticviewer root egasi emas va practiceopsdan chiqarilgan, shuning uchun unga «boshqalar» permissioni qo'llanadi. GUIDE.txtda bu o'qishga (r--) ruxsat beradi. Directoryda esa r-x bor, lekin yozish (w) yo'q. Yangi file yaratish uchun directoryda yozish va o'tish (x) huquqi kerak; shu sabab touch rad etiladi.

8-bosqich: Baholash

```
# Home directoryga qayting.
cd ~
# Baholashdan oldin user nomini tekshiring.
id -un
# P2 bajarilganini baholang.
jdu-check P2
```

Barcha 6 band PASS bo'lsa, M2ga o'ting.

P3 Process va package

8-bobdan keyin 1–4-bosqichlarda processni tekshirib, serviceni to'xtating. 9-bobni o'qib, 5-bosqichda packageni o'rnatish.

O'rganiladigan ko'nikmalar

- systemd service va process PIDini bog'lash.
- Main PIDni ps bilan tekshirish va kerakli serviceni to'xtatish.
- apt bilan package o'rnatish.

Topshiriq

Boshlang'ich holatda jdu-p3-process1.service, jdu-p3-process2.service va jdu-p3-process3.service ishlayapti. figlet packagei o'rnatilmagan. A va Bni bajaring.

Topshiriq A — faqat process2ni to'xtatish. `jdu-p3-process2.service` Main PIDini aniqlang va shu PIDdagi processni `ps` bilan tekshiring. `systemctl stop` orqali serviceni to'xtating. Faqat process2 to'xtasin; process1 va process3 ishlashda davom etsin.

Topshiriq B — package o'rnatish. `figlet` ma'lumotini ko'ring, `apt` bilan o'rnatish va `command`ni bir marta bajaring.

`jdu-check P3 A` va `B`ni bittadan band sifatida baholaydi, jami ikkita. `Command history` yoki yozgan eslatmangiz topshirilmaydi.

Yechim namunasi (qadamma-qadam)

1-bosqich: Uchta serviceni tekshirish

```
# Home directoryga o'ting.
cd ~
# P3 serviselar ro'yxatini ko'rsating.
systemctl list-units --type=service 'jdu-p3-*'
# process1 holatini ko'rsating.
systemctl status jdu-p3-process1.service --no-pager
# process3 holatini ko'rsating.
systemctl status jdu-p3-process3.service --no-pager
```

2-bosqich: Process2 Main PIDini o'qish

```
# process2 holati va Main PID qiymatini ko'rsating.
systemctl status jdu-p3-process2.service
```

Main PID: o'ngidagi raqamni o'zingiz uchun yozib oling. `Pager` ochilsa, `q` bilan chiqing.

3-bosqich: Yozib olingan PIDdagi processni tekshirish

Masalan, yozib olgan raqam 1234 bo'lsa, quyidagi `command`da 1234ni o'z raqamingizga almashtiring:

```
# Yozib olgan PID processini ko'rsating; 1234 o'rniga o'z qiymatingizni qo'ying.
ps -fp 1234
```

`PID`, `USER` va `CMD`ni tekshiring.

4-bosqich: Process2 serviceni to'xtatish

```
# process2 serviceni to'xtating.
sudo systemctl stop jdu-p3-process2.service
# process1 ishlayotganini tekshiring.
systemctl is-active jdu-p3-process1.service
# process2 to'xtaganini tekshiring.
systemctl is-active jdu-p3-process2.service
# process3 ishlayotganini tekshiring.
systemctl is-active jdu-p3-process3.service
```

Tartib bilan `active`, `inactive`, `active` chiqishi kerak.

5-bosqich: Figletni tekshirib o'rnatish

```
# Home directoryga o'ting.
cd ~
# figlet package ma'lumotini o'qing.
apt show figlet
# Mavjud packagelar ro'yxatini yangilang.
sudo apt update
# figletni o'rnatish.
sudo apt install -y figlet
# figlet command joylashuvini tekshiring.
command -v figlet
# figletni bir marta ishga tushiring.
figlet JDU
# P3 bajarilganini baholang.
jdu-check P3
```

figlet JDU yozuv chiqarib, o'zi tugaydi. jdu-check P3 ikkala topshiriqni bittadan band sifatida baholaydi.

Ikkala band PASS bo'lsa, M3ga o'ting.

P4 systemd service

O'rganiladigan ko'nikmalar

- unit fileni o'qish.
- active va enabled holatlarini alohida sozlash.
- Unit sozlamalaridagi User, WorkingDirectory va ExecStartni o'qish.

Topshiriq

jdu-practice-status.service unit filei tayyor, lekin service to'xtagan va avtomatik ishga tushish o'chirilgan. Unit fileni o'zgartirmasdan:

1. User, WorkingDirectory va ExecStartni o'qing.
2. Serviceni ishga tushiring va hozir active ekanini tekshiring.
3. Avtomatik ishga tushishni yoqing va enabled ekanini tekshiring. Bu activedan alohida holat.

jdu-check P4 o'zgarmagan unitning ishlashi va avtomatik ishga tushishini ikkita bandda baholaydi. Kuzatuv natijasini alohida file sifatida topshirmang.

Yechim namunasi (qadamma-qadam)

1-bosqich: Unit fileni o'qish

```
# Home directoryga o'ting.
cd ~
# Service unit sozlamasini ko'rsating.
systemctl cat jdu-practice-status.service
```

User, WorkingDirectory, ExecStartni tekshiring.

2-bosqich: Dastlabki holatni tekshirish

```
# Service hozir ishlayotganini tekshiring.  
systemctl is-active jdu-practice-status.service  
# Boot paytidagi avtomatik ishga tushish yoqilganini tekshiring.  
systemctl is-enabled jdu-practice-status.service
```

3-bosqich: Serviceni start qilish

```
# Serviceni hozir ishga tushiring.  
sudo systemctl start jdu-practice-status.service  
# Servicening joriy holatini tekshiring.  
systemctl is-active jdu-practice-status.service
```

4-bosqich: Tizim yoqilganda start qilishni sozlash

```
# Boot paytidagi avtomatik ishga tushishni yoqing.  
sudo systemctl enable jdu-practice-status.service  
# Avtomatik ishga tushish sozlamasini tekshiring.  
systemctl is-enabled jdu-practice-status.service
```

5-bosqich: Baholash

```
# Home directoryga o'ting.  
cd ~  
# P4 bajarilganini baholang.  
jdu-check P4
```

Ikkala band `PASS` bo'lsa, `M4ga` o'ting.

P5 Port, socket, HTTP, journal

O'rganiladigan ko'nikmalar

- `service`, `process` va `listening socket`ni bog'lash.
- `IP address` va `port`ni o'qish.
- `HTTP request journal`da qayd etilishini tekshirish.

Topshiriq

`jdu-practice-web.service` to'xtagan. `Unit` fileni o'zgartirmang. Quyidagi bog'lanishni yarating va tekshiring:

1. `Unit` filedan `service user`, chiqariladigan `file path`, `IP address` va `port`ni aniqlang. `Serviceni` ishga tushiring.
2. `127.0.0.1:8181`da kutayotgan `socket`ni toping. Uning `PID`ini `service` `Main PID`i bilan solishtiring. Barcha manzillarda kutishga o'zgartirmang.
3. `http://127.0.0.1:8181/ga request` yuborib, `HTTP 200` va javob mazmunini ko'ring. `Service user` `/srv/jdu-practice-web/index.txt`ni o'qiy olishini ham tekshiring.
4. `http://127.0.0.1:8181/p5-checkga` o'zingiz `request` yuboring. `Servicening` joriy ishga tushishiga tegishli `journal`da `REQUEST path=/p5-check` yozuvi paydo bo'lsin.
5. `18181 port`ida `listener` yo'qligini tekshirib, ishlayotgan `8181` bilan solishtiring.

jdu-check P5 to'rtta bandni baholaydi: socket, HTTP va file access, request logi hamda ochiq-yopiq portlar farqi.

Yechim namunasi (qadamma-qadam)

1-bosqich: Unit fileni o'qish

```
# Home directoryga o'ting.  
cd ~  
# Web service unit sozlamasini ko'rsating.  
systemctl cat jdu-practice-web.service
```

127.0.0.1, 8181, jdupracticeweb va kontent filei pathini tekshiring.

2-bosqich: Serviceni start qilish

```
# Web serviceni hozir ishga tushiring.  
sudo systemctl start jdu-practice-web.service  
# Service ishlayotganini tekshiring.  
systemctl is-active jdu-practice-web.service
```

3-bosqich: Listening socketni tekshirish

```
# 8181-portning listening address va PID qiymatini ko'rsating.  
sudo ss -lntp | grep ':8181'
```

127.0.0.1:8181ni toping. Shu chiqishda 0.0.0.0:8181 yoki :::8181da kutish yo'qligini ham tekshiring. Birinchisi faqat shu host ichidan ulanishni, keyingilari barcha manzillarda kutishni bildiradi.

4-bosqich: Socket PID va Main PIDni solishtirish

```
# Service holatini ko'rsating va Main PID raqamini o'qing.  
systemctl status jdu-practice-web.service --no-pager  
# 8181-portni tinglayotgan processning pid= qiymatini o'qing.  
sudo ss -lntp | grep ':8181'
```

systemctl statusdagi Main PID: raqamini ss natijasining users: qatoridagi pid= bilan solishtiring. Teng bo'lsa, servicening asosiy processsi 8181 portida kutmoqda.

5-bosqich: HTTP responseni tekshirish

```
# HTTP response header va body qismlarini ko'rsating.  
curl -i http://127.0.0.1:8181/
```

6-bosqich: Service userning o'qish huquqini tekshirish

Kontent directoriesiga o'ting:

```
# Ochiq file joylashgan directoryga o'ting.
cd /srv/jdu-practice-web
# Joriy directoryni ko'rsating.
pwd
# File yo'lidagi har bir directory permissionini ko'rsating.
ls -ld /srv /srv/jdu-practice-web
# Ochiq filening owner, group va permission qiymatlarini ko'rsating.
ls -l index.txt
# Service user tegishli grouplarni ko'rsating.
id jdupracticeweb
# Service userning login shell qiymatini ko'rsating.
getent passwd jdupracticeweb
# Fileni service user huquqi bilan faqat shu buyruq uchun o'qing.
sudo -u jdupracticeweb -- cat index.txt
```

ls -ld har bir directoryning, ls -l esa index.txtning egasi, groupi va permissionini ko'rsatadi. Fileni o'qish uchun uning read permissioni va har bir yuqori directorydan o'tish (x) huquqi kerak. id service user grouplarini, getent passwd uning login shellini ko'rsatadi.

sudo -u jdupracticeweb -- cat index.txt catni faqat bir marta jdupracticeweb huquqi bilan bajaradi. -- sudo opsiyalari tugaganini bildiradi. Mazmun chiqsa, o'qish mumkin. Bu service userning shelli /usr/sbin/nologin; shuning uchun interaktiv su - ishlatmang.

7-bosqich: Talaba requestini yuborish

```
# Ko'rsatilgan pathga HTTP request yuboring.
curl -i http://127.0.0.1:8181/p5-check
# Servicing so'nggi journal yozuvlarini ko'rsating.
sudo journalctl -u jdu-practice-web.service --no-pager -n 20
```

REQUEST path=/p5-checkni toping.

8-bosqich: Closed port bilan solishtirish

```
# Listening socket yo'q 18181-portga ulanishni sinang.
curl --max-time 2 http://127.0.0.1:18181/
# 18181-portda listening socket yo'qligini tekshiring.
sudo ss -lnt | grep ':18181'
```

Ikkalasi ham muvaffaqiyatli chiqmaydi. TCP 18181da kutayotgan socket yo'q.

9-bosqich: Baholash

```
# Home directoryga o'ting.
cd ~
# P5 bajarilganini baholang.
jdu-check P5
```

To'rttala band PASS bo'lsa, M5ga o'ting.

P6 SSH, remote command, scp

O'rganiladigan ko'nikmalar

- CloudShell va Ubuntuni farqlash.
- scp bilan upload va download qilish.

- SSH remote commandini bajarish.

Topshiriq

CloudShellni ulanish boshlanadigan tomon, Ubuntuni esa uzoqdagi tomon deb oling. SSH sozlamasi va `jdu-ubuntu` ulanish nomi tayyor. P6 filelari hali mavjud emas.

Uzoqdagi serverda yaratiladigan natija filei quyidagi uch satrdan iborat bo'lsin. = o'ngidagi qiymatlarni taxmin qilib qo'lda yozmang; haqiqiy command natijalaridan oling:

```
REMOTE_USER=haqiqiy user nomi
REMOTE_HOST=haqiqiy hostname
REMOTE_PATH=haqiqiy home directory
```

1. CloudShell'dagi `~/jdu-lab/p6/practice-source.txt`ga faqat JDU SSH guided transfer satrini yozing.
2. `scp` bilan uni Ubuntuga `~/jdu-lab/p6/practice-upload.txt` sifatida yuboring. Yuborishdan oldingi va keyingi SHA-256ni solishtiring.
3. CloudShell'dan SSH remote commandni bajaring. Ubuntu haqidagi haqiqiy uch qiymatni yuqoridagi shaklda `~/jdu-lab/p6/practice-remote-result.txt`ga yozing.
4. Ubuntuga ulanib, `jdu-check P6`ni bajaring. Ubuntu tomonidagi ikki band `PASS` bo'lsin.
5. Natija fileini `scp` bilan CloudShellga `~/jdu-lab/p6/practice-downloaded-result.txt` sifatida qaytaring. Ikkala tomondagi SHA-256ni solishtiring.
6. CloudShell'da `jdu-check P6`ni bajaring. CloudShell tomonidagi to'rt band `PASS` bo'lsin.

P6 Ubuntu 2/2 va CloudShell 4/4 bo'lganda tugaydi. O'qituvchi progress sahifasi ham ikki tomonni alohida ko'rsatadi.

Yechim namunasi (qadamma-qadam)

1-bosqich: Ubuntudan CloudShellga qaytish

Ubuntu promptida bajaring:

```
# Ubuntu SSH sessiyasini yakunlab, CloudShellga qayting.
exit
```

CloudShell'da qayerda ekaningizni tekshiring:

```
# CloudShell home directorysiga o'ting.
cd ~
# CloudShell'dagi joriy directoryni ko'rsating.
pwd
# CloudShell user nomini ko'rsating.
id -un
# CloudShell hostname qiymatini ko'rsating.
hostname
```

2-bosqich: CloudShelldagi ish directorysini yaratish

```
# CloudShellda P6 ish directorysini yarating.  
mkdir -p ~/jdu-lab/p6  
# CloudShelldagi P6 directorysiga o'ting.  
cd ~/jdu-lab/p6  
# Joriy directoryni tekshiring.  
pwd
```

3-bosqich: Upload manbasi bo'lgan fileni yaratish

```
# Manba filega ko'rsatilgan bitta satrni yozing.  
printf '%s\n' 'JDU SSH guided transfer' > practice-source.txt  
# Manba file mazmunini ko'rsating.  
cat practice-source.txt  
# Ko'chirishdan oldingi SHA-256 qiymatini ko'rsating.  
sha256sum practice-source.txt
```

4-bosqich: Ubuntudagi directoryni yaratish

CloudShellda bajaring:

```
# SSH orqali Ubuntu tomonidagi P6 directorysini yarating.  
ssh jdu-ubuntu 'mkdir -p ~/jdu-lab/p6'
```

5-bosqich: Ubuntuga upload qilish

CloudShell P6 directorysidan bajaring:

```
# CloudShelldagi P6 directorysiga o'ting.  
cd ~/jdu-lab/p6  
# scp yordamida fileni CloudShelldan Ubuntuga yuboring.  
scp practice-source.txt jdu-ubuntu:~/jdu-lab/p6/practice-upload.txt  
# Ubuntuda qabul qilingan filening SHA-256 qiymatini ko'rsating.  
ssh jdu-ubuntu 'sha256sum ~/jdu-lab/p6/practice-upload.txt'  
# CloudShelldagi manba filening SHA-256 qiymatini ko'rsating.  
sha256sum practice-source.txt
```

Ikki SHA-256 qiymati bir xil bo'lsin.

6-bosqich: SSH remote command bilan natija fileni yaratish

CloudShellda quyidagi bitta commandni aynan ko'rsatilgandek bajaring:

```
# Ubuntudagi uchta haqiqiy qiymatni natija fileiga saqlang.  
ssh jdu-ubuntu 'cd ~/jdu-lab/p6 && printf "REMOTE_USER=%s\nREMOTE_HOST=%s\nREMOTE_PATH=%s\n" "${id -un}"  
"${hostname}" "$HOME" > practice-remote-result.txt'
```

Mazmunni remoteda tekshiring:

```
# SSH orqali Ubuntudagi natija fileni ko'rsating.  
ssh jdu-ubuntu 'cat ~/jdu-lab/p6/practice-remote-result.txt'
```

7-bosqich: Ubuntu tomonini baholash

Ubuntuga ulanib bajaring:

```
# CloudShelldan Ubuntu'ga ulaning.  
ssh jdu-ubuntu  
# Ubuntu'dagi P6 directorysiga o'ting.  
cd ~/jdu-lab/p6  
# Ubuntu'dagi joriy directoryni ko'rsating.  
pwd  
# Ko'chirilgan file va natija fileni tekshiring.  
ls -l  
# Ubuntu tomonidagi P6ni baholang.  
jdu-check P6  
# Ubuntu SSH sessiyasini yakunlab, CloudShellga qayting.  
exit
```

Ubuntu tomondagi ikkala band `PASS` bo'lsin.

8-bosqich: Natija fileni download qilish

CloudShellda bajaring:

```
# CloudShell'dagi P6 directorysiga o'ting.  
cd ~/jdu-lab/p6  
# scp yordamida natija fileni Ubuntu'dan CloudShellga qaytaring.  
scp jdu-ubuntu:~/jdu-lab/p6/practice-remote-result.txt practice-downloaded-result.txt  
# Yuklab olingan file mazmunini ko'rsating.  
cat practice-downloaded-result.txt  
# CloudShell'dagi yuklab olingan filening SHA-256 qiymatini ko'rsating.  
sha256sum practice-downloaded-result.txt  
# Ubuntu'dagi manba filening SHA-256 qiymatini ko'rsating.  
ssh jdu-ubuntu 'sha256sum ~/jdu-lab/p6/practice-remote-result.txt'
```

Ikki SHA-256 qiymati bir xil bo'lsin.

9-bosqich: CloudShell tomonini baholash

```
# CloudShell'dagi P6 directorysiga o'ting.  
cd ~/jdu-lab/p6  
# CloudShell tomonidagi P6ni baholang.  
jdu-check P6
```

CloudShell tomondagi to'rtala band `PASS` bo'lsa, M6ga o'ting.

O'rganish tartibi

Kitobdagi boblar dars soatlari bilan birma-bir mos emas. Tushungan joyingizdan mashqqa o'ting, zarur bo'lsa bobga qayting.

- 1–3-boblar va 4-bobning tahrirlash asoslarini o'qing. Keyin P0 qadamlari bilan haqiqiy OSni kuzating.
- 4–5-boblarni o'qib P1, M1ga o'ting.
- 6–7-boblarni o'qib P2, M2ga o'ting.
- 8-bobni o'qib P3ning 1–4-bosqichlarini bajaring. 9-bobni o'qib P3ning 5-bosqichini bajaring. Keyin M3ga o'ting.
- 10-bobdan keyin P4, M4; 11-bobdan keyin P5, M5ga o'ting.
- 12-bobdan keyin P6, M6ga o'ting. Avvalgi ko'nikmalarni birlashtirib M7ni bajaring.

Umumiy tavsiya etilgan tartib:

```
P0 → P1 → M1 → P2 → M2 → P3 → M3 → P4 → M4  
→ P5 → M5 → P6 → M6 → M7 (yakuniy topshiriq)
```

Pda avval mustaqil urinib, keyin yechimga qarashingiz yoki boshidan namuna bo'yicha ishlashingiz mumkin. M1–M6da P bosqichlariga qaramay, kerakli `command`larni o'zingiz tanlang. M7da M2, M4 va M5dagi bilimlarni birlashtiring.